

## Scenario Overview: Autonomous Amazon Financial Reconciliation Engine

A global enterprise seller processes millions of dollars in monthly revenue across Amazon Seller Central. Every two weeks, Amazon issues raw flat-file settlement reports containing itemized sales, FBA fulfillment fees, advertising costs, tax withholdings, storage adjustments, and promotional discounts.

Currently, finance teams manually reconcile these settlement CSVs against sales orders in **Microsoft Dynamics 365 ERP**, leading to processing delays, unidentified chargebacks, and manual ledger errors.

The objective of the **Day 4 Capstone** is to build an autonomous, production-grade multi-agent engine using the **Microsoft Enterprise Stack** (Microsoft Agent Framework, Microsoft Foundry Services, Azure Services, Dynamics 365, and GitHub). The engine automatically ingests raw settlement files, parses line items, executes 2-way/3-way deterministic matching against Dynamics 365 AR, resolves fee variances using AI reasoning, routes high-value exceptions to human managers, and posts finalized general journal entries with OpenTelemetry audit tracking.

## Phase-Specific Governance Rules

### Phase 1: Environment & Platform Provisioning Rules

1. **Network & Identity Boundary Rule:** All Azure resources (Storage Accounts, Foundry Hubs, Logic Apps) must operate within isolated VNet perimeters with private endpoints. System-Assigned Managed Identities must be used exclusively—hardcoded credentials, static secrets, or API keys are strictly forbidden.
2. **Access Control Least-Privilege Rule:** Agent roles must be granted granular RBAC rights (e.g., Storage Blob Data Contributor). Unscoped subscription-level contributor rights are prohibited.

3. **Connector Standardization Rule:** Platform-to-ERP communication must route through official Azure Logic Apps connectors using Entra ID authentication. Custom HTTP wrappers or external API gateways are not allowed.

## **Phase 2: Multi-Agent System Implementation Rules**

4. **Deterministic Precedence Rule:** Agents must execute a 2-way/3-way deterministic lookup against Dynamics 365 AR before calling any AI reasoning model. LLM processing is permitted only for unresolved variances, promotional discounts, or missing SKU mappings.
5. **Schema Fail-Fast Rule:** Incoming Amazon settlement files must undergo instant structural schema validation. Invalid or corrupted files must route immediately to /quarantine without incurring LLM token costs.
6. **Idempotent Posting Rule:** The ERP Posting Agent must perform a lookup on the unique Amazon SettlementID before issuing ledger entries to prevent duplicate double-posting.

## **Phase 3: Safety Guardrails & Human-in-the-Loop Setup Rules**

7. **PII Masking Rule:** All settlement text payloads must pass through Microsoft Foundry Control Plane Content Safety filters to redact customer names, addresses, and phone numbers before being processed by an LLM.
8. **Hard Financial Variance Ceiling Rule:** Any file or line-item variance exceeding **\$500 OR 5%** of the net total must pause automated processing and emit a human approval task to the Microsoft Foundry Agent Service HITL Canvas. Agents cannot self-approve items above this limit.

## **Phase 4: Observability, Evaluation & Cloud Deployment Rules**

9. **Full Lineage Tracing Rule:** Every agent execution, state transition, and tool call must emit OpenTelemetry-compliant spans directly to the Microsoft Foundry Control Plane Observability Engine.
10. **Evaluation Quality Gate Rule:** Agent deployments must achieve a minimum threshold score of **95% Grounding** and **98% Reconciliation Accuracy** on synthetic benchmark suites prior to production release.

**Phase 5: Live Execution & Handoff Rules**

11. **Zero Unaudited Posting Rule:** No ledger entry may be posted to Dynamics 365 without an accompanying OpenTelemetry trace ID and an immutable transaction log entry.
12. **State Recovery Rule:** In the event of an ERP network interruption, agents must persist state and resume execution without losing historical audit lineage or re-triggering completed sub-tasks.

**Sample Amazon Settlement Data (10 Test Invoices)**

Below are 10 structured settlement line items representing typical Amazon flat-file data used for engine ingestion and testing:

| Invoice / Order ID | Amazon ASIN | Line Type | Item Amount | FBA Fee / Deduction | Promo / Tax            | Net Amount | Scenario Type        |
|--------------------|-------------|-----------|-------------|---------------------|------------------------|------------|----------------------|
| INV-AMZ-001        | B08N5WRWNW  | Order     | \$150.00    | - \$22.50 (FBA Fee) | -\$5.00 (Tax Withheld) | \$122.50   | Standard 3-Way Match |

| <b>Invo<br/>ice /<br/>Ord<br/>er ID</b> | <b>Amazon<br/>ASIN</b> | <b>Line<br/>Type</b> | <b>Item<br/>Amou<br/>nt</b> | <b>FBA<br/>Fee /<br/>Deduc<br/>tion</b>             | <b>Promo /<br/>Tax</b>              | <b>Net<br/>Amo<br/>unt</b> | <b>Scenari<br/>o Type</b>                    |
|-----------------------------------------|------------------------|----------------------|-----------------------------|-----------------------------------------------------|-------------------------------------|----------------------------|----------------------------------------------|
| <b>INV-<br/>AMZ<br/>-002</b>            | B07XJ8C8<br>F5         | Orde<br>r            | \$450.<br>00                | -<br>\$67.50<br>(FBA<br>Fee)                        | -\$15.00<br>(Promo)                 | \$367.<br>50               | Unalloc<br>ated<br>Promo<br>(Smart<br>Match) |
| <b>INV-<br/>AMZ<br/>-003</b>            | B09B2W8<br>S7E         | Refu<br>nd           | -<br>\$120.<br>00           | +\$18.0<br>0<br>(Resto<br>ck<br>Credit)             | +\$4.00<br>(Tax<br>Refund)          | -<br>\$98.0<br>0           | Custom<br>er<br>Return<br>Process<br>ing     |
| <b>INV-<br/>AMZ<br/>-004</b>            | B01M102<br>O2D         | Orde<br>r            | \$1,20<br>0.00              | -<br>\$180.0<br>0 (FBA<br>Fee)                      | -\$80.00<br>(Fee<br>Adjust<br>ment) | \$940.<br>00               | High<br>Varianc<br>e<br>(Trigger<br>s HITL)  |
| <b>INV-<br/>AMZ<br/>-005</b>            | B0813X28<br>KP         | Orde<br>r            | \$85.0<br>0                 | -<br>\$12.75<br>(FBA<br>Fee)                        | \$0.00                              | \$72.2<br>5                | Standar<br>d 2-Way<br>Match                  |
| <b>INV-<br/>AMZ<br/>-006</b>            | B07PVF6L<br>29         | Stor<br>age          | \$0.00                      | -<br>\$620.0<br>0<br>(Long-<br>Term<br>Storag<br>e) | \$0.00                              | -<br>\$620.<br>00          | Fee<br>Overch<br>arge<br>(Trigger<br>s HITL) |

| <b>Invo<br/>ice /<br/>Ord<br/>er ID</b> | <b>Amazon<br/>ASIN</b> | <b>Line<br/>Type</b> | <b>Item<br/>Amou<br/>nt</b> | <b>FBA<br/>Fee /<br/>Deduc<br/>tion</b> | <b>Promo /<br/>Tax</b>           | <b>Net<br/>Amo<br/>unt</b> | <b>Scenari<br/>o Type</b>                                  |
|-----------------------------------------|------------------------|----------------------|-----------------------------|-----------------------------------------|----------------------------------|----------------------------|------------------------------------------------------------|
| <b>INV-<br/>AMZ<br/>-007</b>            | B09H21KL<br>09         | Orde<br>r            | €250.<br>00                 | -<br>€37.50<br>(FBA<br>Fee)             | €0.00                            | €212.<br>50                | Currenc<br>y<br>Conver<br>sion<br>(EUR to<br>USD)          |
| <b>INV-<br/>AMZ<br/>-008</b>            | INVALID-<br>ID         | Orde<br>r            | \$200.<br>00                | -<br>\$30.00<br>(FBA<br>Fee)            | \$0.00                           | \$170.<br>00               | Schem<br>a /<br>Validati<br>on<br>Failure                  |
| <b>INV-<br/>AMZ<br/>-009</b>            | B08N5WR<br>WNW         | Orde<br>r            | \$150.<br>00                | -<br>\$22.50<br>(FBA<br>Fee)            | -\$5.00<br>(Tax<br>Withhel<br>d) | \$122.<br>50               | Duplica<br>te<br>Transac<br>tion ID<br>Test                |
| <b>INV-<br/>AMZ<br/>-010</b>            | B031K9L0<br>2A         | Orde<br>r            | \$310.<br>00                | -<br>\$46.50<br>(FBA<br>Fee)            | -\$10.00<br>(Promo)              | \$253.<br>50               | Bounda<br>ry<br>Varianc<br>e (<<br>\$500<br>thresho<br>ld) |

#### **Verification & Test Suite (20 Test Cases)**

To verify the capstone deployment, all 20 test cases must be executed and pass validation:

## 1. Ingestion & Validation

- **TC-01 (Valid File Trigger):** Upload standard Amazon CSV to /raw-settlements → Confirm Ingestion Agent triggers and processes header rows.
- **TC-02 (Corrupted Schema Routing):** Upload CSV with missing mandatory headers → Confirm file moves to /quarantine and run halts with zero LLM execution.
- **TC-03 (Large File Chunking):** Ingest a 50,000-row settlement CSV → Verify streaming ingestion handles file without memory limits or timeout exceptions.

## 2. Extraction & Parsing

- **TC-04 (Line-Item Extraction Accuracy):** Compare extracted JSON against raw CSV for 100 line items → Confirm 100% field precision across ASIN, order ID, and fee columns.
- **TC-05 (Negative Adjustment Allocation):** Input refund and restock fee line items → Verify correct ledger balance signs in output JSON.
- **TC-06 (Multi-Currency Parsing):** Process European EUR settlement lines → Verify agent extracts correct currency code without converting values prematurely.

## 3. Deterministic & Smart Matching

- **TC-07 (Exact 3-Way Match):** Input exact matching order ID, amount, and open AR balance → Verify 100% automated match without calling smart reasoning model.

- **TC-08 (Fuzzy Promo Code Matching):** Input order with unallocated \$15 promotional discount → Verify Smart Matching Agent grounds on Amazon policy and allocates variance.
- **TC-09 (FX Rate Resolution):** Process EUR line item against USD Dynamics 365 AR account → Verify agent queries system exchange rates and matches balance within 0.1% tolerance.
- **TC-10 (Unmatched SKU Routing):** Submit an order ID with a non-existent Dynamics 365 SKU → Verify system flags item as "Unmapped Product" and isolates record.

#### 4. Governance, Safety & HITL

- **TC-11 (PII Redaction):** Ingest customer delivery note containing name and phone number → Verify Microsoft Foundry Content Safety redacts PII before model processing.
- **TC-12 (Dollar Variance HITL Trigger):** Process invoice with a \$620 storage fee discrepancy → Confirm state shifts to PENDING\_HUMAN\_APPROVAL and appears in Foundry Canvas.
- **TC-13 (Percentage Variance HITL Trigger):** Process invoice with an 8% fee variance (\$80 on \$1,000 order) → Confirm HITL trigger fires due to percentage threshold violation.
- **TC-14 (HITL Approval Execution):** Click "Approve" on a paused HITL task in Foundry Canvas → Confirm execution resumes and passes record to ERP Posting Agent.
- **TC-15 (HITL Rejection Workflow):** Click "Reject" on an invalid variance item → Confirm record routes to manual finance queue and halts posting.

## 5. ERP Integration & Anti-Double Posting

- **TC-16 (Idempotency Anti-Double Post):** Re-submit an already processed SettlementID → Confirm lookup flags duplicate and blocks redundant journal entry.
- **TC-17 (ERP Posting Success):** Submit fully reconciled journal payload → Verify entry creates successfully in Dynamics 365 General Journal via Logic Apps.

## 6. Observability, Security & Performance

- **TC-18 (Zero-Trust Identity Validation):** Revoke static secrets → Verify pipeline continues executing using Managed Identities.
- **TC-19 (OpenTelemetry Span Verification):** Execute end-to-end reconciliation → Verify full trace span (Ingestion → Smart Match → ERP Post) appears in Foundry Control Plane.
- **TC-20 (State Rehydration Failure Recovery):** Simulate network failure mid-transaction → Verify system rehydrates state from last persistent checkpoint and resumes without data loss.



To complete the **Day 4 Capstone: Autonomous End-to-End Amazon Settlement & Reconciliation Engine**, the implementation guide is updated below to incorporate all phase-specific governance rules, test data, and technical requirements.

## Step-by-Step Capstone Execution Guide

### Phase 1: Infrastructure & Pipeline Setup

- **Step 1.1: Provision Azure Isolated Infrastructure via Azure Portal UI**
  - Sign in to the Azure Portal using your enterprise tenant credentials.
  - **Configure Private Virtual Network:** Create an Azure VNet (vnet-amazon-prod) with dedicated subnets for integration and private endpoints.
  - **Provision Storage Account:**
    - Create Storage Account stdamazonreconcile (Standard LRS).
    - Under *Networking*, set connectivity to **Disabled** for public access and configure a **Private Endpoint** mapped to your VNet's private endpoint subnet for blob storage.
    - Under *Data Management > Containers*, add three private containers: /raw-settlements, /processed-json, and /quarantine.
  - **Provision Foundry Resources:**

- Navigate to Microsoft Foundry and deploy a Hub instance along with Project instance (foundry-amazon-prod).
- Under *Networking* for the Foundry Project, set network access to **Private with Private Endpoint** and attach it to your VNet perimeter.

- **Step 1.2: Configure Entra ID & Least-Privilege RBAC**

- Enable **System-Assigned Managed Identity** across all Azure Services and the GitHub Runner. *Static credentials and API keys are strictly forbidden.*
- Assign granular RBAC roles using the principle of least privilege:
  - Assign Storage Blob Data Contributor to the Foundry Project identity.
  - Assign Dynamics 365 Business Central/Finance Contributor to the Azure Logic App Managed Identity.

- **Step 1.3: Bind Standardized Dynamics 365 Connectors**

- Create an Azure Logic App workflow within the private network perimeter.

- Connect directly to Dynamics 365 ERP using the official **Azure Logic Apps Dynamics 365 Connector** authenticated strictly via Entra ID Managed Identity.
- Define HTTP webhook endpoints inside Logic Apps to allow downstream agents to query Accounts Receivable (AR) balances and post general journal entries.

## Phase 2: Multi-Agent Development

- **Step 2.1: Construct the Ingestion Agent (Schema Fail-Fast Rule)**

- Initialize a Python environment using azure-ai-projects and microsoft-agent-framework SDKs.
- Configure an Azure Event Grid blob trigger that fires when a settlement CSV lands in /raw-settlements.
- **Schema Fail-Fast Validation:** Before calling any LLM API, programmatically check headers for mandatory Amazon fields (order-id, sku, price-type, amount).
- *Action on Error (e.g., INV-AMZ-008):* Immediately transfer the file to /quarantine and terminate the pipeline without consuming LLM tokens.

- **Step 2.2: Build the Remittance Extraction Agent**

- Define Pydantic / JSON Schema structures to map flat-file settlement records into normalized JSON.

- Extract line-item breakdowns and maintain precision across line types: Gross Sales, FBA Fulfillment Fees, Storage Fees, Amazon Advertising, Withholding Taxes, and Net Settlement Amount.
  - Support negative adjustments (such as INV-AMZ-003 refund restock credits) and maintain raw currency codes (e.g., EUR for INV-AMZ-007) without early conversion.
- **Step 2.3: Program Payment & Deterministic Smart Matching Agents**
    - **Deterministic Loop First (2-Way / 3-Way Match):** Execute an explicit C# or Python lookup loop that matches extracted order-id values directly against open Dynamics 365 AR Sales Orders via the Logic Apps API before making any AI call. (Applies to standard orders like INV-AMZ-001 and INV-AMZ-005).
    - **Smart Reasoning Loop:** Invoke an LLM agent grounded on Amazon Vendor Guidelines stored in Microsoft Foundry IQ *only* for unmatched lines, FX rate resolutions (INV-AMZ-007), or unallocated promotional discounts (INV-AMZ-002).
- **Step 2.4: Build the ERP Posting Agent (Idempotence & Validation)**
    - Implement pre-posting validation to ensure  $\text{Net Settlement} = \text{Total Sales} - \text{Total Fees} \pm \text{Adjustments}$ .
    - **Idempotent Posting Lookup:** Before issuing any ledger entries, check the unique SettlementID against Dynamics 365 to prevent double-posting (e.g., catching duplicate submission INV-AMZ-009).

- Stage structured journal entry payloads ready for Dynamics 365 ledger entry.

### **Phase 3: Governance, Guardrails & HITL Implementation**

- **Step 3.1: Enforce Content Safety & PII Redaction**

- Pass all incoming settlement text payloads through Microsoft Foundry Control Plane Content Safety filters.
- Run regex and Microsoft Presidio/Foundry PII classifiers to redact customer names, shipping addresses, and phone numbers from transaction notes prior to model reasoning.

- **Step 3.2: Program HITL Threshold Evaluation (Hard Financial Variance Ceiling)**

- Insert a decision node within the agent state graph that calculates variance using the following formula:

$$\text{Variance \%} = \left( \frac{|\text{Calculated Settlement} - \text{Amazon Reported Settlement}|}{\text{Amazon Reported Settlement}} \right) \times 100$$

- Evaluate incoming records against the threshold rules:
  - If Variance Amount > \$500 (e.g., fee adjustment on INV-AMZ-004 or storage charge on INV-AMZ-006), **OR**
  - If Variance % > 5% of net total,

- Then set execution state to PENDING\_HUMAN\_APPROVAL. Agents cannot self-approve items exceeding these ceilings. (Note: Boundary variances like INV-AMZ-010 under \$500/5% proceed automatically).
- **Step 3.3: Configure Foundry Agent Service HITL Canvas**
  - Route PENDING\_HUMAN\_APPROVAL events to a RequestPort that generates an approval task in the Microsoft Foundry Agent Service HITL Canvas.
  - Pause agent state execution until a Finance Manager approves or rejects the variance breakdown.
  - Resume state processing upon approval and pass the verified transaction payload to the ERP Posting Agent.

## **Phase 4: Observability, Testing & CI/CD Deployment**

- **Step 4.1: Integrate OpenTelemetry Distributed Tracing (Full Lineage)**

- Instrument all agent functions with OpenTelemetry spans:

Python

```
from opentelemetry import trace
```

```
tracer = trace.get_tracer("amazon.reconciliation")
```

```
with tracer.start_as_current_span("smart_matching_execution"):
```

# Deterministic check -> Smart match logic

pass

- Direct trace exporters to the Microsoft Foundry Control Plane Observability Engine to record trace IDs, latency, execution steps, and token consumption.

- **Step 4.2: Run Evaluation Suites via Foundry Evaluators (Quality Gates)**

- Execute a test suite containing 50 synthetic Amazon settlement files representing diverse edge cases.
- Enforce pre-deployment evaluation gates via Foundry Evaluators: require minimum scores of **95% Grounding** and **98% Reconciliation Accuracy**.

- **Step 4.3: Automate Production Release via GitHub Actions**

- Deploy using the deploy-agents.yml workflow.
- Include automated red-teaming/jailbreak tests within the pipeline.
- Package agent runtimes into serverless container instances hosted on Microsoft Foundry Hosted Agents.

## **Phase 5: Live Execution, Validation & Handoff**

- **Step 5.1: Execute Live End-to-End Test**

- Drop a multi-currency flat file into /raw-settlements.
  - Validate that INV-AMZ-008 immediately routes to /quarantine (Schema Fail-Fast) while valid files trigger extraction automatically.
  - Confirm deterministic matching executes for standard line items (INV-AMZ-001, INV-AMZ-005), and Smart Matching handles FX conversions (INV-AMZ-007) and promos (INV-AMZ-002).
- **Step 5.2: Demonstrate HITL, Zero-Trust, and ERP Finalization**
    - Process high-variance items (INV-AMZ-004 and INV-AMZ-006) to verify they halt execution and trigger tasks in the Foundry HITL Canvas.
    - Approve the task in the HITL Canvas and ensure the ERP Posting Agent posts the journal entry to Dynamics 365, complete with an attached OpenTelemetry trace ID.
    - Attempt re-submitting INV-AMZ-009 to confirm the idempotency check blocks double-posting.
- **Step 5.3: Verify Observability & State Recovery**
    - Review the Microsoft Foundry Control Plane Dashboard for:
      - Automated Reconciliation Rate (Straight-Through Processing %).



- Complete OpenTelemetry traces (Ingestion → Smart Match → ERP Post).
- Mid-transaction recovery confirmation (simulating an ERP interruption to verify state rehydration without duplicate processing).
- Deliver repository runbooks and hand over the environment.

#### Verification Test Suite Alignment Matrix

| Test Case | Category   | Input Scenario                  | Expected Outcome                                                       |
|-----------|------------|---------------------------------|------------------------------------------------------------------------|
| TC-01     | Ingestion  | Upload standard CSV             | Ingestion Agent triggers via Event Grid and parses headers.            |
| TC-02     | Ingestion  | INV-AMZ-008 (Invalid Header/ID) | Schema Fail-Fast triggers; routes to /quarantine with \$0 token usage. |
| TC-03     | Ingestion  | 50,000-row CSV                  | Streaming ingestion completes without memory or timeout errors.        |
| TC-04     | Extraction | 100 Line items                  | 100% precision across ASIN, order ID, and fee breakdowns.              |
| TC-05     | Extraction | INV-AMZ-003 (Refund/Restock)    | Output JSON contains correct ledger balance signs (−\$98.00).          |

| Test Case | Category   | Input Scenario                  | Expected Outcome                                                     |
|-----------|------------|---------------------------------|----------------------------------------------------------------------|
| TC-06     | Extraction | INV-AMZ-007 (EUR Currency)      | Extracted line preserves raw currency code (EUR) prior to matching.  |
| TC-07     | Matching   | INV-AMZ-001 / INV-AMZ-005       | Deterministic 2/3-way match succeeds without LLM call.               |
| TC-08     | Matching   | INV-AMZ-002 (Unallocated Promo) | Smart Matching Agent applies Amazon guidelines to allocate variance. |
| TC-09     | Matching   | INV-AMZ-007 (EUR to USD)        | System retrieves FX rates and matches within 0.1% tolerance.         |
| TC-10     | Matching   | Unmapped SKU                    | Flags item as "Unmapped Product" and isolates the record.            |
| TC-11     | Safety     | Delivery note with PII          | Presidio/Content Safety redacts PII before model processing.         |
| TC-12     | HITL       | INV-AMZ-006 (\$620 Storage Fee) | Halts execution (> \$500) and sends task to Foundry Canvas.          |
| TC-13     | HITL       | INV-AMZ-004 (8% Fee Variance)   | Halts execution (> 5%) and sends task to Foundry Canvas.             |
| TC-14     | HITL       | Approved HITL Task              | Execution resumes and payload moves to ERP Posting Agent.            |

| <b>Test Case</b> | <b>Category</b> | <b>Input Scenario</b>         | <b>Expected Outcome</b>                                             |
|------------------|-----------------|-------------------------------|---------------------------------------------------------------------|
| <b>TC-15</b>     | HITL            | Rejected HITL Task            | Item routes to manual finance queue; ERP posting is suppressed.     |
| <b>TC-16</b>     | ERP Post        | INV-AMZ-009<br>(Duplicate ID) | Idempotency check detects existing ID and blocks posting.           |
| <b>TC-17</b>     | ERP Post        | Reconciled Journal Payload    | Journal entry posts to Dynamics 365 via Logic App connector.        |
| <b>TC-18</b>     | Security        | Revoked Static Secrets        | System executes using Managed Identity authentication.              |
| <b>TC-19</b>     | Observability   | Full Reconciliation Run       | Complete OpenTelemetry trace span appears in Foundry Control Plane. |
| <b>TC-20</b>     | Performance     | Network Disruption Mid-Run    | System rehydrates state from checkpoint and resumes execution.      |

# Autonomous End-to-end Amazon Settlement & Reconciliation Engine

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